

```

1  import random
2  import string
3
4  def generate_password(length):
5      # Define character sets
6      letters = string.ascii_letters
7      digits = string.digits
8      special_chars = string.punctuation
9
10     # Combine all characters
11     all_chars = letters + digits + special_chars
12
13     # Ensure password contains at least one of each type
14     password = [
15         random.choice(letters),
16         random.choice(digits),
17         random.choice(special_chars)
18     ]
19
20     # Fill remaining length
21     password += random.choices(all_chars, k=length - 3)
22
23     # Shuffle the password
24     random.shuffle(password)
25
26     # Convert list to string
27     return ''.join(password)
28
29 # Main program
30 try:
31     length = int(input("Enter password length: "))
32
33     if length < 3:
34         print("Password length should be at least 3")
35     else:
36         password = generate_password(length)
37         print("Generated Password:", password)
38
39 except ValueError:
40     print("Please enter a valid number!")

```